



FactorMoE: a framework with mixture-of-experts and attention to dynamically combine alpha factors in quantitative investment

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Abstract

Financial markets are characterized by high volatility and low signal-to-noise ratios, placing stringent demands on quantitative investment methods. Formulaic alpha factors are valued for their interpretability but often lack adaptivity, whereas machine learning-based alpha factors deliver stronger predictive performance at the expense of transparency. Here we introduce FactorMoE, a deep-learning framework for the dynamic combination of formulaic alpha factors that reconciles this interpretability-performance trade-off. FactorMoE conditions learnable gating networks on market regime indicators and recent factor performance to dynamically activate and weight candidate alpha factors. A chained Mixture-of-Experts architecture coupled with Multi-Head Attention enables sequential expert refinement and the modelling of nonlinear, synergistic interactions among factors, allowing the framework to prioritize robust factor combinations as market conditions evolve. Empirical evaluation on recent real market data shows that the proposed framework outperforms contemporary state-of-the-art and baseline methods on both factor-selection and stock-picking tasks, validating its effectiveness and superiority. The code is available at the <https://github.com/ZCJ66q/FactorMoE> repository.

Keywords Alpha factors · Quantitative investment · Dynamic combination · FactorMoE

Introduction

In quantitative trading, an alpha factor is a quantifiable signal extracted from market or firm-level data that is used to predict an asset's future excess returns [21]. Because alpha factors are closely related to stock trends, identifying factors that deliver high excess returns has long been an active topic for investors and researchers [9]. Modern methods to alpha factor discovery can be broadly classified into two categories: machine learning methods and formulaic alpha methods. Traditional formulaic alpha factors, such as the Fama-French three-factor [6] and five-factor [7] models, construct signals based on economic theory and statistical rules and therefore offer strong interpretability. However, these factors are static, making them ill-suited to accommodate high-frequency fluctuations and changes in market structure; moreover, once widely adopted, their predictive power often decays rapidly [3]. Otherwise, many

deep-learning approaches rely on base architectures such as LSTM [10] and Transformer [26] models. For example, MASTER [15] proposes a market-guided stock price prediction model. These models can typically generate more complex alpha factors and automatically capture nonlinearities and higher-order interaction effects. However, their "black-box" nature reduces the transparency of the resulting strategies. To balance interpretability and adaptability, researchers have turned to automated factor-discovery methods. Early genetic programming algorithm [16] were used to automate the discovery of formulaic alpha factors. However, their limited search space and representational capacity hindered the discovery of more complex alphas exhibiting synergistic effects. More recently, AlphaForge [24] proposed a two-stage framework for alpha factor generation and combination: the generation stage leverages the expansive search capabilities of deep learning to efficiently produce formulaic alphas that are lowly correlated and highly adaptive, while the combination stage employs a dynamic linear-combination model to adjust factor weights in real time.

In recent years, linear combination methods for alpha factors have been widely adopted in quantitative investing due to their simplicity and strong interpretability, and they

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tend to perform well when market conditions are stable. However, as market structures grow increasingly complex and data dimensionality rises, traditional linear combination approaches have revealed limitations in expressive capacity and struggle to capture nonlinear relationships [12]. Factor screening procedures commonly depend heavily on historical metrics, which makes it difficult to capture complex nonlinear interactions and synergistic effects among alpha factors in turbulent markets. For instance, during extreme market events or structural regime shifts, some alpha factors may exhibit markedly divergent performance; static screening schemes and conventional linear weighting often fail to accommodate such changes, resulting in delayed responses and degraded portfolio performance.

To address the issues above, we propose the FactorMoE, a framework inspired by the two-stage architecture of AlphaForge [24]. FactorMoE reconciles the interpretability of formulaic alpha factors with the high adaptability of machine learning-based factors. Its core components are a Chained Mixture-of-Experts (CoE) [27], Multi-Head Attention [26], Scoring Experts, and dynamic gating mechanism. The framework jointly analyzes market regime indicators and alpha factor performances to compute composite scores for individual factors, applies these scores to determine factor importance, and subsequently aggregates the selected factors via a linear projection. Compared with traditional static screening and linear weighting, FactorMoE can activate the most suitable factor combinations in real time, substantially improving sensitivity to market shifts and enhancing predictive performance.

To evaluate FactorMoE's alpha factor combination performance, we conducted a series of experiments. The results show that FactorMoE preserves the dynamic adaptability and strong predictive capability of alpha factor combination under both normal and extreme market conditions, and achieves higher returns in simulated trading. The main contributions of this paper are summarized as follows:

- Our framework incorporates deep-learning modules that enable it to discover alpha factor combinations with enhanced adaptability and stronger synergistic interactions in complex market conditions.
- Our framework introduces a dynamic gating mechanism. Unlike traditional static selection and linear combination approaches, our method incorporates a novel score-weighting step that enables the dynamic activation of alpha factors.
- We performed a comprehensive suite of experiments to validate the proposed framework and compared it against baseline and state-of-the-art models on contemporary datasets. The results demonstrate the superiority of our approach.

Preliminary

Alpha factor

Consider a trading period of length T , denote the total number of stocks by N . On each trading day $t \in \{1, 2, \dots, T\}$, the feature vector for stock n is $x_{tn} \in \mathbb{R}^{m\tau}$, which represents m raw features computed over a rolling window of the past τ trading day. This paper uses six raw features: open, close, low, high, volume, and vwap. We define an alpha factor as a mapping function f that maps the collection of all stocks' feature vectors on day t , denoted $X_t \in \mathbb{R}^{N \times m\tau}$, to alpha values $v_t = f(X_t) \in \mathbb{R}^N$. Hereafter, the inputs to the framework are the alpha values.

Alpha factor combination

An alpha factor pool of size K obtained from formulaic or machine-learning approaches. The feature matrix on day t , $X_t \in \mathbb{R}^{N \times m\tau}$, is mapped to alpha values for all K factors as $v_{Kt} = f_{1,\dots,K}(X_t) \in \mathbb{R}^{N \times K}$, where $f_{1,\dots,K}$ denotes the set of K mapping functions corresponding to the different alpha factors. We then compute $S(v_{Kt})$, where the mapping S denotes an evaluation function for alpha factors. Based on the evaluation results, we select a subset of alpha factors with superior combined performance and, by exploiting complementarities among factors, reduce the risk of any single factor—thereby improving the portfolio's robustness and return profile [8].

Mixture of experts

A mixture-of-experts (MoE) model is a machine learning architecture that dynamically assigns multiple expert subnetworks to process different aspects of the input. Let the total number of experts be I with $i \in \{1, 2, \dots, I\}$. Denote the mapping implemented by expert i as E_i and the gating-network mapping as G . The core idea of the MoE is to route an input x to the most relevant expert subnetworks, producing expert outputs $E_i(x)$. the final output y is obtained as the weighted sum of the expert outputs using the gating weights $G(x)_i$ [23]:

$$y = G(x)_1 E_1(x) + G(x)_2 E_2(x) + \dots + G(x)_I E_I(x) \quad (1)$$

Methodology

Our framework consists of four components: (A) The Formulaic Alpha Factor Mining module, responsible for producing a set of candidate formulaic alpha factors. (B) Market-Guided Attention CoE (MACoE). MACoE computes market-guided alpha factor scores via three components—a CoE, a Multi-

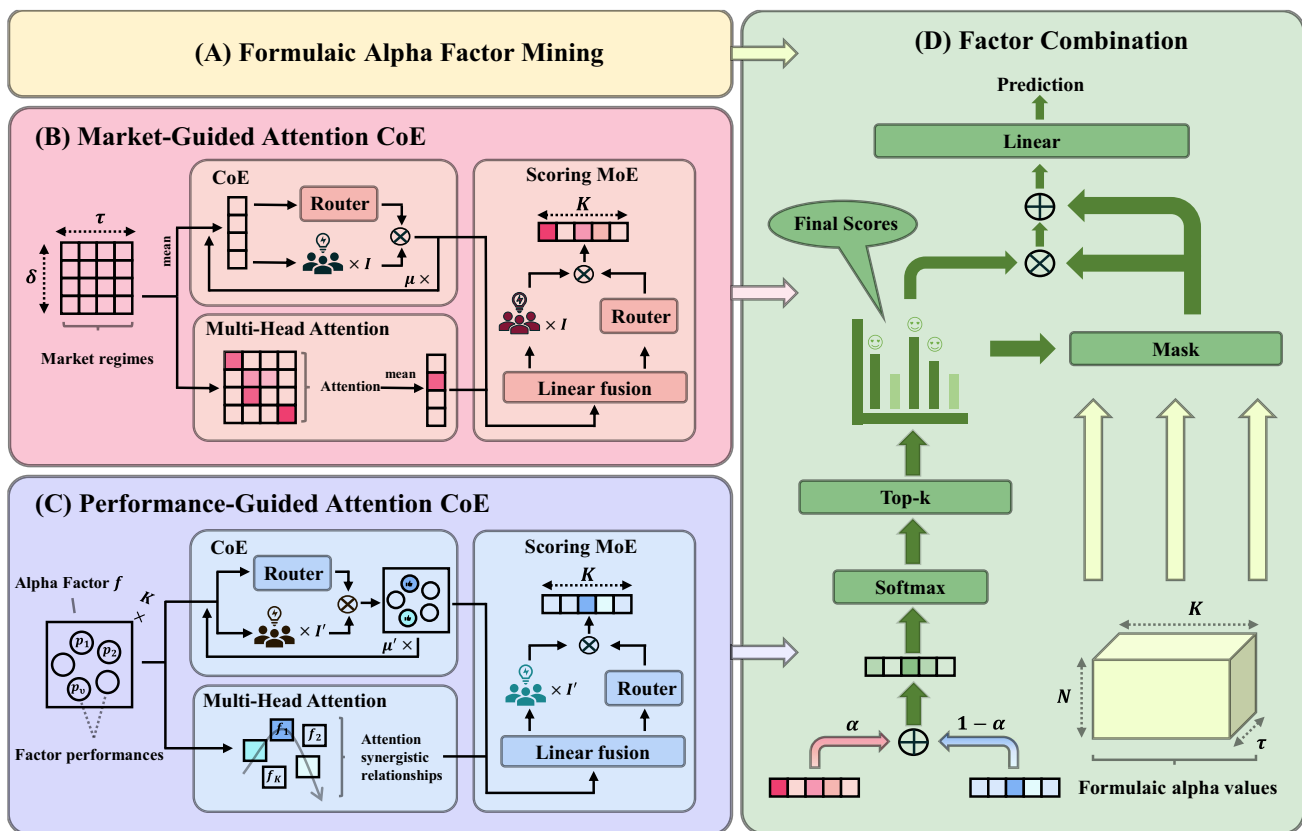


Fig. 1 Overview of the FactorMoE framework. **A** The alpha factor generation module. **B** The proposed market-guided alpha factor scoring module. **C** The proposed performance-guided alpha factor scoring

module. **D** The proposed alpha factor dynamic-activation gating network. Panels **B–D** together form the alpha factor dynamic combination module

Head Attention, and an a Scoring MoE. (C) Performance-Guided Attention CoE (PACoE). Factors that have performed well in the past tend to exhibit superior performance in the future [4]. However, the predictive power of alpha factors may weaken over time or even reverse direction. To address this, we design a PACoE that preferentially selects recently strong factors with favorable synergies. This module likewise comprises a CoE, a Multi-Head Attention, and a Scoring MoE. (D) Factor Combination. This module dynamically aggregates market scores and alpha factor performance scores and uses the final scores to select alpha factors adaptively. Figure 1 illustrates the proposed FactorMoE framework. The following subsections provide a detailed description of FactorMoE.

Market-guided attention CoE

Part (B) of Fig. 1 illustrates the detailed architecture of the MACoE module. We first construct a vector that can describe market regime indicators, including engineered technical indicators [22], market breadth and sentiment features [13]. Engineered technical indicators are extracted from the mar-

ket's stock price and trading volume data through feature engineering methods, such as the moving averages of closing prices for individual stocks in the market over different periods. Market breadth and sentiment features are statistical indicators that reflect the overall market's price movements, trading activity, and changes in investor sentiment. They are used to characterize the market's internal structure and sentiment state, such as the number of stocks with price increases and decreases in the daily market. Partition vector into $\gamma \in \mathbb{R}^{\tau \times \delta}$ by rolling window, where τ denotes the rolling window size and δ denotes the number of market regime indicators.

We feed γ into the Multi-Head Attention module. In this component, γ is processed by a multi-head attention operator $\text{MHA}(\cdot)$ followed by a linear projection. The resulting features are averaged along the τ dimension to yield γ_1 . This module captures complex inter-temporal dependencies in the market and effectively makes up for the temporal features that may be omitted by the CoE.

Expert specialization, where different experts focus on distinct tasks, reduces interference and facilitates more efficient learning [5]. Accordingly, the expert pool in this

#	Algorithm 1: Market-Guided Attention CoE Pipeline
1	Input: market regime indicators $\gamma \in \mathbb{R}^{\tau \times \delta}$
2	Output: the set of alpha factor scores obtained under market regime guidance γ_4
3	Multi-Head Attention: $\#W_*^1, b_{MHA}^1$ are learnable matrix and bias
4	$Q_\gamma^1 \leftarrow W_Q^1 \gamma, K_\gamma^1 \leftarrow W_K^1 \gamma, V_\gamma^1 \leftarrow W_V^1 \gamma$
5	$\gamma_1 \in \mathbb{R}^{1 \times \delta} \leftarrow \text{Mean}((W_{MHA}^1 \times \text{MHA}^1(Q_\gamma^1, K_\gamma^1, V_\gamma^1) + b_{MHA}^1), \text{dim} = \tau)$
6	CoE: $\#W_1$ is a learnable matrix
7	$\gamma_{input} \in \mathbb{R}^{1 \times \delta} \leftarrow \text{Mean}(\gamma, \text{dim} = \tau)$ and initialize γ_2
8	for μ iteration do
9	$W_2 \leftarrow \text{Softmax}_{\beta_1}(W_1 \times \gamma_{input})$
10	$\gamma_2 \in \mathbb{R}^{1 \times \delta} \leftarrow W_2 \times \text{Stack}[E_{m,1}(\gamma_{input}), \dots, E_{m,I}(\gamma_{input})]$ and $\gamma_{input} \leftarrow \gamma_2$
11	end for
12	Scoring MoE: $\#W_3, b_1$ are learnable matrix and bias
13	$\gamma_3 \in \mathbb{R}^{1 \times \delta} \leftarrow W_3 \times (\gamma_1 \gamma_2) + b_1$ and $W_4 \leftarrow \text{Softmax}_{\beta_1}(W_1 \times \gamma_3)$
14	$\gamma_4 \in \mathbb{R}^{1 \times K} \leftarrow W_4 \times \text{Stack}[E'_{m,1}(\gamma_3), \dots, E'_{m,I}(\gamma_3)]$
15	return γ_4

component comprises two groups: market regimes experts and scoring experts. Both groups share an identical internal architecture: two Linear layers, a ReLU [1] activation, and a Dropout layer.

In the CoE module, the γ_{input} is processed by a Router to produce expert weights $W_2 \in \mathbb{R}^{1 \times I}$, which are scaled by the coefficient β_1 . Where I denotes the number of experts and β_1 is a scaling coefficient. Smaller values of β_1 concentrate attention on individual experts. Concurrently, γ_{input} is fed in parallel to each expert E_m . Every expert nonlinearly maps the input via a hidden dimension ε , and the resulting expert outputs are aggregated using the expert weights. After μ iterations, the final result is assigned to γ_2 . By emulating hierarchical abstraction and progressive feature extraction, the CoE captures complex nonlinear relationships, while its multi-round chained architecture enhances adaptability in complex market regimes.

To obtain a more comprehensive and detailed representation of market regimes and to improve the robustness of the alpha factor scoring, we fuse γ_1 and γ_2 with a single linear layer. Where $||$ denotes the concatenation of vectors. Each scoring expert E'_m then maps the fused representation γ_3 through a hidden dimension ε to produce factor scores. Finally, the expert outputs are aggregated using the expert-assignment weights to yield γ_4 . To improve stability, the scaling coefficient β_1 and the number of experts in this stage are set equal to those used in the CoE module. Algorithm 1 provides a detailed description of the MACoE model.

Performance-guided attention CoE

Part (C) of Fig. 1 depicts the PACoE module. We evaluate alpha factor performance using a rolling window. Specifically, the performance measures comprise: (1) the

#	Algorithm 2: Performance-Guided Attention CoE Pipeline
1	Input: alpha factor performances $\phi \in \mathbb{R}^{K \times v}$
2	Output: the set of alpha factor scores obtained under performance guidance ϕ_4
3	Multi-Head Attention: $\#W_*^2, b_{MHA}^2$ are learnable matrix and bias
4	$Q_\phi^2 \leftarrow W_Q^2 \phi, K_\phi^2 \leftarrow W_K^2 \phi, V_\phi^2 \leftarrow W_V^2 \phi$
5	$\phi_1 \in \mathbb{R}^{K \times v} \leftarrow W_{MHA}^2 \times \text{MHA}^2(Q_\phi^2, K_\phi^2, V_\phi^2) + b_{MHA}^2$
6	CoE: $\#W_5$ is a learnable matrix
7	$\phi_{input} \leftarrow \phi$ and initialize ϕ_2
8	for μ' iteration do
9	$W_6 \leftarrow \text{Softmax}_{\beta_2}(W_5 \times \phi_{input})$
10	$\phi_2 \in \mathbb{R}^{K \times v} \leftarrow W_6 \times \text{Stack}[E_{p,1}(\phi_{input}), \dots, E_{p,I'}(\phi_{input})]$ and $\phi_{input} \leftarrow \phi_2$
11	end for
12	Scoring MoE: $\#W_7, b_2$ are learnable matrix and bias
13	$\phi_3 \in \mathbb{R}^{K \times v} \leftarrow W_7 \times (\phi_1 \phi_2) + b_2$ and $W_8 \leftarrow \text{Softmax}_{\beta_2}(W_5 \times \phi_3)$
14	$\phi_4 \in \mathbb{R}^{1 \times K} \leftarrow W_8 \times \text{Stack}[E'_{p,1}(\phi_3), \dots, E'_{p,I'}(\phi_3)]$
15	return ϕ_4

Information Coefficient (IC), defined as the Pearson correlation between an alpha factor and the target within the window. IC provides an effective measure of predictive power. (2) the Rank Information Coefficient (RIC), defined as the Spearman rank correlation between the alpha factor and the target within the window. RIC evaluates ranking performance. (3) the cross-sectional regression slope [20], which quantifies the change in the target associated with a one-unit change in the alpha value.

We first construct, for each alpha factor, a performance vector computed over a rolling window. We then average the performance vectors along the rolling window dimension to obtain a compact summary that represents each window's performance. We denote this representative vector by $\phi \in \mathbb{R}^{K \times v}$, where K is the number of alpha factors and v is the number of performances per factor. The vector ϕ is then fed concurrently into the CoE module and the Multi-Head Attention module.

By leveraging the attention mechanism within the Multi-Head Attention module, the model can discover latent alpha factor combinations that exhibit synergistic interactions and selectively attend to higher-performing factors.

In the CoE module, each expert E_p applies diverse nonlinear mapping to the performance vector of every alpha factor via a hidden dimension ϵ . Each expert implements a distinct feature processing function, and their outputs are aggregated by weighting them with expert weights $W_6 \in \mathbb{R}^{K \times I'}$. Where K and I' denote the number of alpha factors and the number of experts, respectively, and β_2 is the scaling coefficient applied to the expert weights. After μ' iterations, the final result is assigned to ϕ_2 . The chained architecture of the CoE enables the successive flow and fusion of information across

experts, progressively refining richer representations of factor performance.

We fuse ϕ_1 and ϕ_2 to obtain ϕ_3 . The scoring experts E'_p then project ϕ_3 into the hidden dimension ϵ to generate factor scores, which are aggregated using the expert weights W_8 . The model's final output is the performance-guided score vector for the alpha factors, denoted ϕ_4 . Algorithm 2 provides a detailed description of the PACoE model.

Factor combination

Previous studies have adopted voting-based techniques that perform static fusion by aggregating the predictions of multiple classifiers [18], where all classifiers are treated as equally active. In contrast, our module leverages the adaptive inputs from MACoE and PACoE to dynamically combine alpha factors by jointly considering market regime indicators and alpha factor performance. As shown in part (D) of Fig. 1, the inputs are the alpha factor scores derived from market regime indicators γ_4 , and the scores derived from alpha factor performance ϕ_4 . These two score vectors are dynamically fused via a learnable parameter α :

$$\delta = \text{Softmax} \left[\frac{\alpha \times \gamma_4 + (1 - \alpha) \times \phi_4}{\beta_3} \right] \quad (2)$$

where $\delta \in \mathbb{R}^{1 \times K}$, α is initialized to 0.5. β_3 is the scaling coefficient: smaller values of β_3 produce sparser fusion scores in δ , while larger values yield denser allocations, thereby providing control over the concentration of the score distribution.

Prior to Top-k selection, we perturb the fused score vector δ with Gumbel [14] noise. This step increases randomness in the combination process, facilitating exploration of potentially effective alpha factor combinations during early training. We then apply Top-k [28] operation to the perturbed scores to select the k most representative factors (with $k \leq K$), and construct a corresponding one-hot Mask. δ is subsequently masked and normalized to produce the normalized fusion vector $\delta_1 \in \mathbb{R}^{1 \times K}$:

$$\delta_1 = \frac{\text{Top-k}(\delta) \times \text{Mask}(\delta)}{\sum_{i=0}^K [\text{Top-k}(\delta) \times \text{Mask}(\delta)]_i} \quad (3)$$

Next, δ_1 is expanded to match the dimensionality of the alpha values vector $v \in \mathbb{R}^{T \times K \times N}$ and applied as a selector to realize dynamic combination of alpha factors. Simultaneously, a residual connection is applied to the masked alpha values:

$$v' = \delta_1 \times v + \text{mask}(\delta) \times v \quad (4)$$

Finally, the output v' is passed to the predictor $P(\cdot)$ to perform label regression. The predictor is implemented as a single linear layer, and prediction quality is assessed by the MSE loss. At each rolling window, the learnable parameters are updated by gradient-based optimization via backpropagation using the MSE loss between the framework output $\hat{Y}$ and the label Y .

$$\hat{Y} = P(v'), \quad \text{Loss} = \text{MSE}(\hat{Y}, Y) \quad (5)$$

Experiment

The following questions were addressed through a series of experiments:

Q1: Does our framework outperform recent formulaic methods, contemporary machine-learning approaches, and baseline models?

Q2: How does performance vary with the size of the alpha factor pool?

Q3: Are the individual components of our framework effective?

Q4: How robust is the framework under more realistic trading conditions?

Experiment settings

Datasets

We evaluate our framework on the CSI300, CSI500 and CSI800 datasets. The CSI300 index comprises the 300 largest and most liquid stocks listed on the Shanghai and Shenzhen exchanges, whereas the CSI500 covers 500 smaller-capitalization, higher-growth stocks; together they provide a representative and diverse testbed. The CSI800 index is formed by the constituents of the CSI300 and CSI500 and therefore reflects the overall trend of stocks in the A-share market. We split datasets into training, validation, and test sets covering 2012-01-01 to 2022-12-31, 2023-01-01 to 2023-12-31, and 2024-01-01 to 2024-12-31, respectively. The stock data is publicly available and was downloaded using Qlib [29]. We define the label as "Ref(close, -20)/close - 1", namely, the return over the 20 trading days following the current close. We devise 54 engineered technical indicator features [22] and 18 market breadth and sentiment features. In addition, we compute three summary statistics of alpha factor performance—mean, standard deviation, and the mean-to-standard-deviation ratio—yielding 9 performance features per factor.

Compared methods

To evaluate our framework, we compare its alpha combination performance against several representative approaches. XGBoost [2]: a decision-tree based method that effectively captures nonlinear relationships in financial markets. LSTM [10]: a recurrent neural network capable of modeling long-term dependencies, widely applied to financial time-series forecasting. Transformer [26]: an encoder-decoder architecture based on self-attention; its attention mechanism can effectively capture salient patterns in market data. Mamba [11]: a sequence model that dynamically stores and retrieves salient information and, owing to its near-linear computational complexity, is well suited for large-scale financial data analysis. MASTER [15]: a state-of-the-art market-guided stock price prediction model. To validate the generality of our approach, we also evaluate it using alpha factor pools produced by several advanced formulaic mining approaches. AlphaGen [30] (RL): an approach based on Reinforcement Learning; AlphaAgent [25] (LLMs): an approach based on Large Language Models; AlphaForge [24] (GANs): an approach based on generative adversarial networks.

Evaluation metrics

We evaluate model performance using both ranking and portfolio-based metrics. The ranking metrics comprise the Information Coefficient (IC), the Rank Information Coefficient (RankIC), the Information Ratio based IC (ICIR) and the Information Ratio based RankIC (RankICIR). ICIR and RankICIR are computed as the mean of IC (or RankIC) divided by its standard deviation. These ranking metrics have been used previously to assess model performance [15, 24]. In addition, we report two portfolio-level metrics to compare return and risk across methods. We simulate a simple daily trading rule that buys the top 10% of stocks by predicted return and compute Excess Annualized Return (AR) and the Information Ratio (IR). AR is defined as the strategy's annualized return minus the benchmark's annualized return (the corresponding index), and IR is defined as the mean of AR divided by its standard deviation, serving as a measure of return stability.

Training details

We implemented FactorMoE with Pytorch [19]. The rolling window length τ is set to 40. For the market component, the hidden dimension ε is tuned from {128,256}; the number of experts I is 8; the expert chain depth μ is 4; the number of attention heads is tuned from {3,4}; and β_1 is 0.5. For the performance component, the hidden dimension ϵ is 32; the number of experts I' is 2; the expert chain depth μ' is 2; the number of attention heads is 3; and β_2 is 2. In the alpha fac-

tor combination component, β_3 is 3. The number of selected factors is set between 0.6 and 1.0 of the total alpha factor pool. The initial learning rate is 2×10^{-4} . Training epochs is capped at 20 epochs with early stopping. The primitive operator sets and fitness functions for the underlying mining models follow the standard configurations described in AlphaGen [30], AlphaForge [24] and AlphaAgent [25].

Backtesting setup

Alpha factors were constructed using six raw features—open, close, low, high, volume, and volume-weighted average price (vwap)—to predict asset returns. The portfolio construction strategy imposed a maximum holding limit of 50 equities, with a mandatory minimum holding period of 20 trading days per asset. Subject to these constraints, the top five stocks ranked by the model's predictive signal were acquired daily, while the five lowest-ranked stocks within the existing portfolio were liquidated. Simulations were conducted over the period from January 1, 2023, to December 1, 2025, utilizing the CSI 500. Transaction costs were modeled at 5 basis points for both buy and sell executions.

Overall performance

We feed the computed alpha values to the compared methods and contrast their outputs with those produced by our proposed approach. Table 1 reports the results for formulaic alpha factors mined using GANs across different combination strategies. Results using other mining methods follow the same pattern and are omitted for brevity. As shown in Tables 1 and 2, our method achieves superior performance across multiple metrics, demonstrating strong factor combination capability. Moreover, the approach generalizes well across different alpha mining procedures. In particular, the model exhibits strong stock-selection performance on ranking metrics such as IC and RankIC, and even under a simple trading rule delivers excellent portfolio-level results. This subsection answers Q1.

Effect of the pool size

Since ranking metrics more directly reflect a model's stock-selection capability, whereas portfolio-based metrics are heavily influenced by the specifics of the trading strategy, we restrict the experiments in this section to ranking metrics only. To answer Q2, we evaluate our framework using alpha factor pools of sizes {5,10,20,50,100}. For each pool size, we mine five distinct sets of alpha factors, run experiments on each set, and report the mean performance across these five trials. As shown in Fig. 2, the relationship between performance and pool size is not monotonic. Because our method dynamically weights factors using both

Table 1 Performance comparison of different combination methods on CSI300, CSI500 and CSI800

Dataset	Model	IC	ICIR	RankIC	RankICIR	AR	IR
CSI300	XGBoost	0.030	0.260	0.042	0.313	<u>0.112</u>	<u>2.08</u>
	LSTM	0.045	<u>0.331</u>	0.055	<u>0.372</u>	0.099	1.95
	Transformer	<u>0.047</u>	0.280	<u>0.063</u>	0.346	0.059	1.29
	Mamba	0.038	0.326	0.046	0.346	0.077	1.47
	MASTER	0.034	0.272	0.045	0.338	0.063	1.30
	Ours	0.061	0.374	0.077	0.443	0.135	2.65
CSI500	XGBoost	0.009	0.101	0.030	0.285	0.075	1.12
	LSTM	<u>0.042</u>	0.308	<u>0.056</u>	0.395	0.096	1.39
	Transformer	0.039	0.285	0.052	0.344	<u>0.121</u>	<u>1.91</u>
	Mamba	0.037	<u>0.343</u>	0.052	<u>0.401</u>	0.118	1.89
	MASTER	0.041	0.316	0.051	0.367	0.099	1.42
	Ours	0.054	0.382	0.081	0.514	0.139	2.08
CSI800	XGBoost	0.014	0.170	0.021	0.211	<u>0.057</u>	1.00
	LSTM	<u>0.035</u>	<u>0.246</u>	<u>0.048</u>	<u>0.319</u>	0.011	0.17
	Transformer	0.021	0.165	0.031	0.235	0.040	0.60
	Mamba	0.021	0.232	0.025	0.239	0.035	0.56
	MASTER	0.024	0.178	0.045	0.308	0.054	0.78
	Ours	0.040	0.306	0.051	0.350	0.059	<u>0.92</u>

The best results are in bold and the second-best results are underlined

Table 2 Combination performance of our method across different formulaic alpha factor mining models

Dataset	Model	IC	ICIR	RankIC	RankICIR	AR	IR
CSI300	RL	0.044	0.266	0.047	0.273	0.106	1.85
	RL+Ours	<u>0.055</u>	<u>0.340</u>	<u>0.060</u>	<u>0.348</u>	0.134	<u>2.39</u>
	LLMs	0.033	0.216	0.028	0.184	0.111	1.57
	LLMs+Ours	0.035	0.222	0.031	0.202	0.150	2.07
	GANs	0.041	0.296	0.048	0.316	0.119	2.28
	GANs+Ours	0.061	0.374	0.077	0.443	<u>0.135</u>	2.65
CSI500	RL	0.015	0.120	0.016	0.125	0.019	0.27
	RL+Ours	0.029	0.206	<u>0.038</u>	<u>0.272</u>	0.050	0.77
	LLMs	0.038	0.331	0.028	0.209	0.079	1.11
	LLMs+Ours	<u>0.040</u>	<u>0.349</u>	0.032	0.227	<u>0.098</u>	<u>1.35</u>
	GANs	0.024	0.178	0.037	0.266	0.070	1.12
	GANs+Ours	0.054	0.382	0.081	0.514	0.139	2.08
CSI800	RL	0.029	0.247	0.045	0.369	0.053	0.90
	RL+Ours	0.033	<u>0.285</u>	0.051	0.404	<u>0.066</u>	1.12
	LLMs	0.034	0.268	0.023	0.163	0.047	0.67
	LLMs+Ours	<u>0.036</u>	0.275	0.025	0.175	0.070	<u>0.97</u>
	GANs	0.015	0.126	0.025	0.175	0.032	0.51
	GANs+Ours	0.040	0.306	<u>0.051</u>	<u>0.350</u>	0.059	0.92

“+Ours” denotes results obtained when our method is used to perform the combination. The best results are in bold and the second-best results are underlined

market regime indicators and recent factor performance, it can reliably select highly synergistic and robust alpha factors even in high-volatility market conditions. Consequently, the method exhibits strong stability as the pool size varies

and maintains competitive performance even with small factor pools—an attractive property for practical deployments with constrained resources.

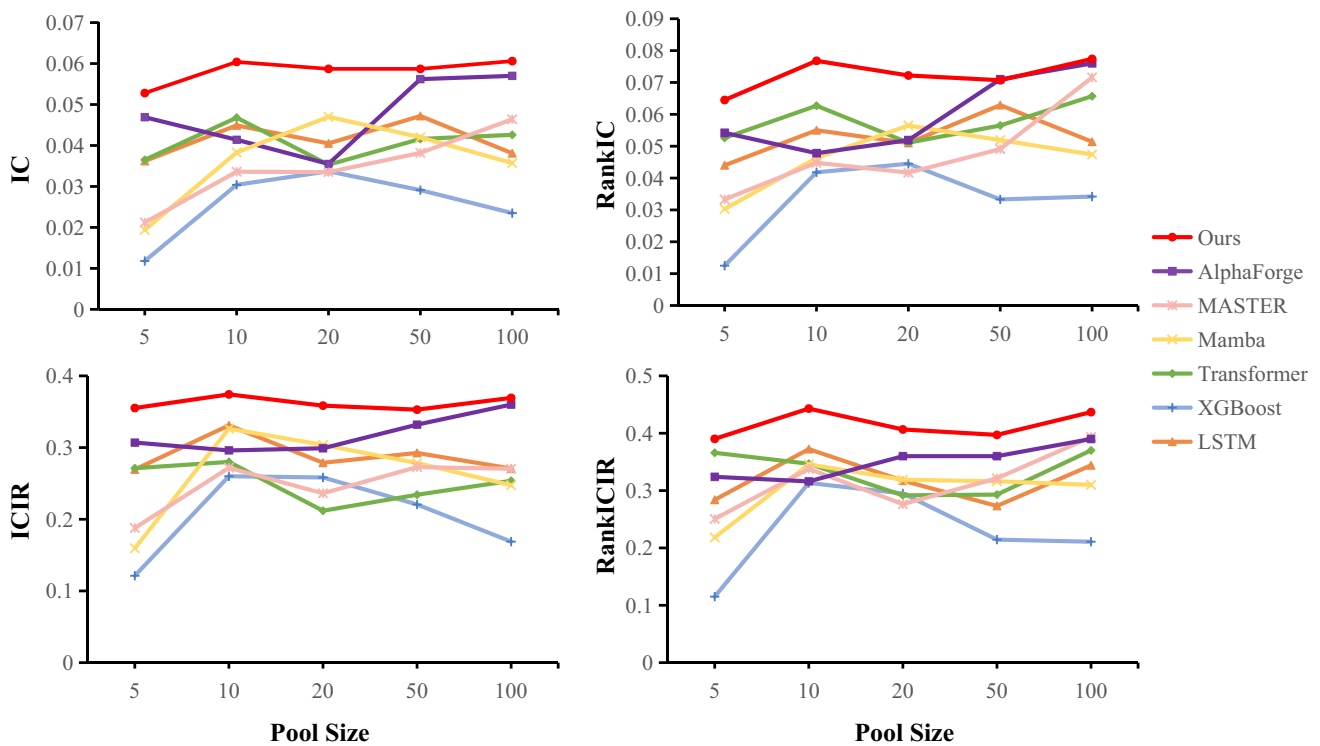


Fig. 2 Comparison of results between our framework and other methods on CSI300 across Different Pool Size

Table 3 Experiments on CSI300 to validate the effectiveness of the proposed framework

Model	IC	ICIR	RankIC	RankICIR
FactorMoE	0.0587	0.3585	0.0722	0.4066
Linear	0.0478	0.2940	0.0625	0.3661
-Market	0.0544	0.3351	0.0682	0.3846
-Performance	0.0573	0.3484	0.0711	0.3981
-MCoE	<u>0.0576</u>	<u>0.3534</u>	<u>0.0718</u>	0.4072
-PCoE	0.0553	0.3381	0.0692	0.3914
-MAtten	0.0530	0.3228	0.0667	0.3725
-PAtten	0.0553	0.3348	0.0685	0.3859
-MSMoE	0.0538	0.3320	0.0681	0.3850
-PSMoE	0.0551	0.3455	0.0666	0.3797

The best results are in bold and the second-best results are underlined

Ablation study

As in the previous subsection, we restrict the experimental evaluation in this subsection to ranking metrics only. We validate the contribution of each component of the FactorMoE framework through 9 ablation settings: (1) Linear—alpha factors are combined exclusively via linear weighting; (2) -Market—the MACoE component is removed; (3) -Performance—the PACoE component is removed; (4) -MCoE—the CoE submodule is removed from the MACoE;

(5) -PCoE—the CoE submodule is removed from the PACoE; (6) -MAtten—the Multi-Head Attention submodule is removed from the MACoE; (7) -PAtten—the Multi-Head Attention submodule is removed from the PACoE; (8) -MSMoE—the scoring MoE submodule is removed from the MACoE; and (9) -PSMoE—the scoring MoE submodule is removed from the PACoE.

Table 3 presents the ablation results for a pool size of 20. To address **Q3**, we performed a series of experiments in which one component of FactorMoE was removed at a time. The results show that each component contributes positively to overall performance, and the model achieves its best results when all components are retained.

Table 4 reports the impact of hyperparameter configurations on framework performance. The Default setting corresponds to our experimental configuration: rolling window size $\tau = 40$; MACoE hidden dimension $\varepsilon = 256$; number of market experts $I = 8$; expert chain depth $\mu = 4$; number of attention heads $h = 4$; and scaling coefficient $\beta_1 = 0.5$. For the PACoE, the hidden dimension $\epsilon = 32$; number of experts $I' = 2$; chain depth $\mu' = 2$; number of attention heads $h' = 3$; and scaling coefficient $\beta_2 = 2$. The factor pool size is 20, and the number of selected factors k is set to 0.8 of the total, i.e., 16. The results indicate that the relationship between individual hyperparameters and overall performance is nonlinear, with the number of experts exerting a particularly pronounced influence. Overall, the adopted

Table 4 Impact of hyperparameter configurations on framework performance on the CSI300

Hyperparameters	IC	ICIR	RankIC	RankICIR
Default	0.0587	0.3585	0.0722	0.4066
$I = 4, I' = 1$	0.0482	0.2984	0.0625	0.3572
$I = 16, I' = 4$	0.0462	0.2899	0.0588	0.3440
$\mu = 2, \mu' = 1$	0.0558	0.3416	0.0685	0.3866
$\mu = 8, \mu' = 4$	0.0552	0.3338	0.0691	0.3857
$\beta_1 = 2, \beta_2 = 2$	0.0575	0.3524	0.0718	0.4038
$\beta_1 = 4, \beta_2 = 4$	0.0567	0.3480	0.0712	0.3998
$\varepsilon = 128, \epsilon = 16$	0.0511	0.3181	0.0634	0.3610
$\varepsilon = 512, \epsilon = 64$	0.0540	0.3325	0.0663	0.3687
$h = 2, h' = 1$	0.0578	0.3520	0.0715	0.4015
$h = 6, h' = 3$	0.0565	0.3455	0.0705	0.3976
$k = 10$	0.0554	0.3508	0.0683	0.3913
$k = 20$	0.0552	0.3353	0.0701	0.3974
$\tau = 30$	0.0561	0.3467	0.0717	0.4082
$\tau = 50$	0.0548	0.3307	0.0703	0.3900

“Default” denotes the well-tuned configuration adopted in our experiments, while the remaining entries report performance after adjusting individual hyperparameters. The best results are in bold

default configuration achieves a well-balanced and competitive performance.

Case study

As shown in Table 5, we present a simple case that illustrates our framework’s alpha factor combination capability. The table reports 10 alpha factors selected from a pool of 20. Notably, selections can differ even on adjacent days, and individual factor scores evolve over time.

For example, item 6, “Corr(high,volume,10)”, measures the correlation between a stock’s daily high price and its trading volume over the preceding 10 trading days. This alpha factor captures the short-term degree to which volume accompanies price spikes and therefore serves as an indicator of market sentiment. Its relatively high score indicates that the model regards it as important. A negative weight assigned to this factor may be interpreted economically as signaling that when price declines are accompanied by panic-driven selling, the security could represent an attractive buying opportunity.

The learned parameter α indicates that the model places greater emphasis on market information than on alpha fac-

Table 5 Usage of alpha factors on two consecutive days

#	Exprs(Parameter $\alpha = 0.64$)	Score	Weight
2	Corr((((vwap*(close*10.0))*-5.0)/-30.0)*high)/0.5,volume,40)	0.0027	0.0003
3	Max((S_log1p((S_log1p((1.0*((Corr(volume,(30.0+vwap),10)-5.0)+5.0)))-10.0))*-2.0),50)	0.0237	0.0027
4	Sum(((0.5*(high+10.0))*-0.5),40)	0.0209	-0.0004
6	Corr(high,volume,10)	0.1644	-0.0038
9	((Inv(close)/1.0)*Inv(S_log1p((Mean((-1.0/close),30)+30.0)-30.0)))	0.0071	-0.0066
11	(S_log1p(Min((Inv((Min((S_log1p((2.0-close))+0.01),20)*-0.5)*-0.01))/-1.0),30))+5.0)	0.0169	0.0018
13	Std(S_log1p(S_log1p(Sum((-30.0-((volume*0.01)-30.0)),30))),30)	0.2383	-0.0003
14	Std(S_log1p(S_log1p((-30.0/(((volume*-0.5)-5.0)+-5.0))),30)	0.0229	0.0002
15	(Std(S_log1p(S_log1p(Sum((-30.0+((volume-0.5)+30.0)),20))),20)-30.0)	0.1103	0.0005
20	Std(((0.5*((vwap*open)-5.0))+Inv(((0.01**Delta(ts_div(close,50),20))-30.0))),10)	0.3925	-0.0015
#	Exprs(Parameter $\alpha = 0.64$)	Score	Weight
2	Corr((((vwap*(close*10.0))*-5.0)/-30.0)*high)/0.5,volume,40)	0.0027	0.0003
3	Max((S_log1p((S_log1p((1.0*((Corr(volume,(30.0+vwap),10)-5.0)+5.0)))-10.0))*-2.0),50)	0.0237	0.0027
4	Sum(((0.5*(high+10.0))*-0.5),40)	0.0209	-0.0004
6	Corr(high,volume,10)	0.1638	-0.0038
11	(S_log1p(Min((Inv((Min((S_log1p((2.0-close))+0.01),20)*-0.5)*-0.01))/-1.0),30))+5.0)	0.0169	0.0018
13	Std(S_log1p(S_log1p(Sum((-30.0-((volume*0.01)-30.0)),30))),30)	0.2374	-0.0003
14	Std(S_log1p(S_log1p((-30.0/(((volume*-0.5)-5.0)+-5.0))),30)	0.0227	0.0002
15	(Std(S_log1p(S_log1p(Sum((-30.0+((volume-0.5)+30.0)),20))),20)-30.0)	0.1099	0.0005
18	(S_log1p(Inv(Inv(Std(S_log1p((S_log1p((Min((vwap/0.01),40)/30.0))/0.5)*5.0)),20))))+5.0)	0.0107	0.0048
20	Std(((0.5*((vwap*open)-5.0))+Inv(((0.01**Delta(ts_div(close,50),20))-30.0))),10)	0.3911	-0.0015

The upper table corresponds to Day 1 and the lower table to Day 2. Exprs denote the formulaic alpha factors employed on that day. The parameter α is learnable and is used to aggregate scores. Score indicates the final score, and Weight denotes the weight assigned by the Predictor

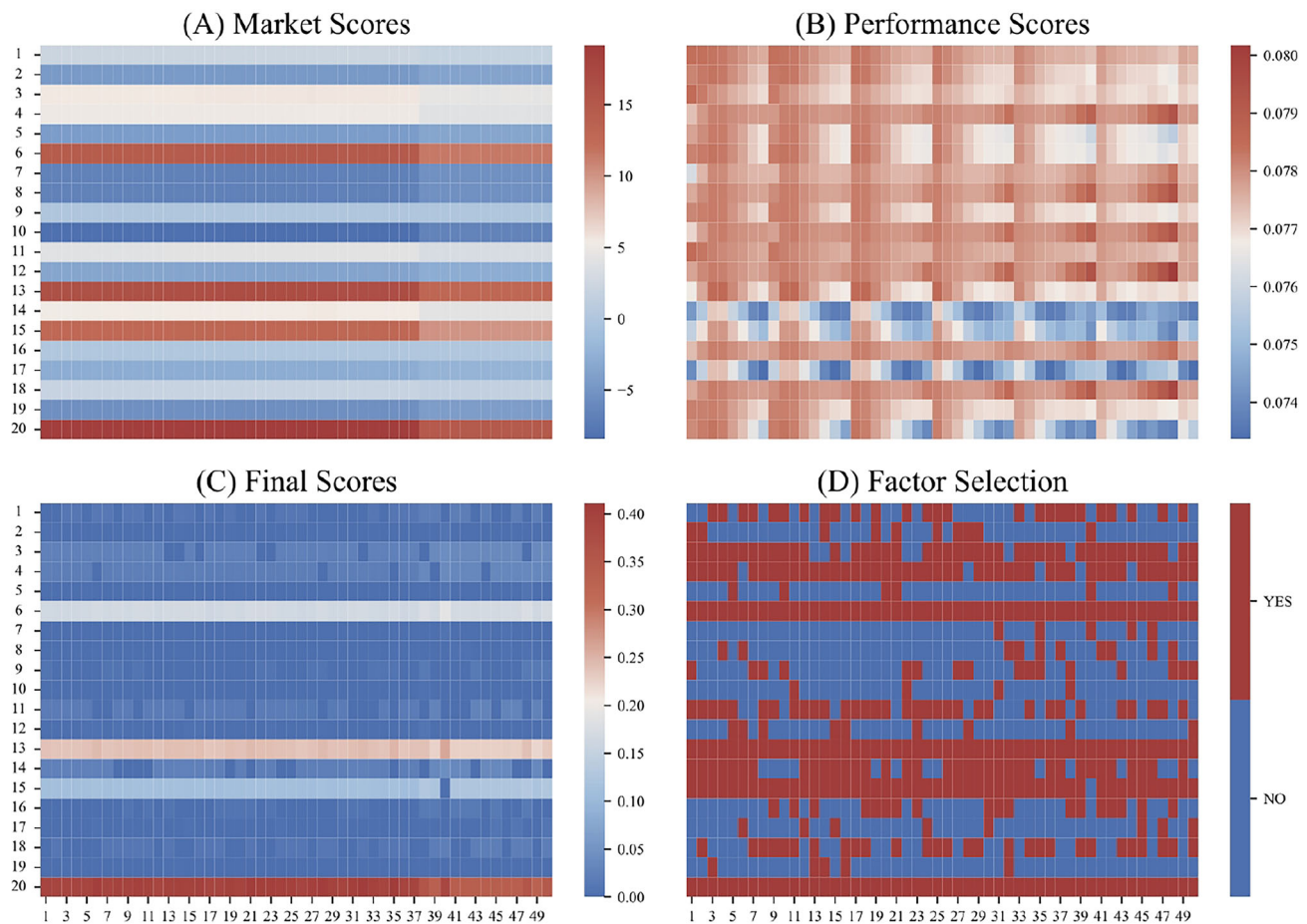
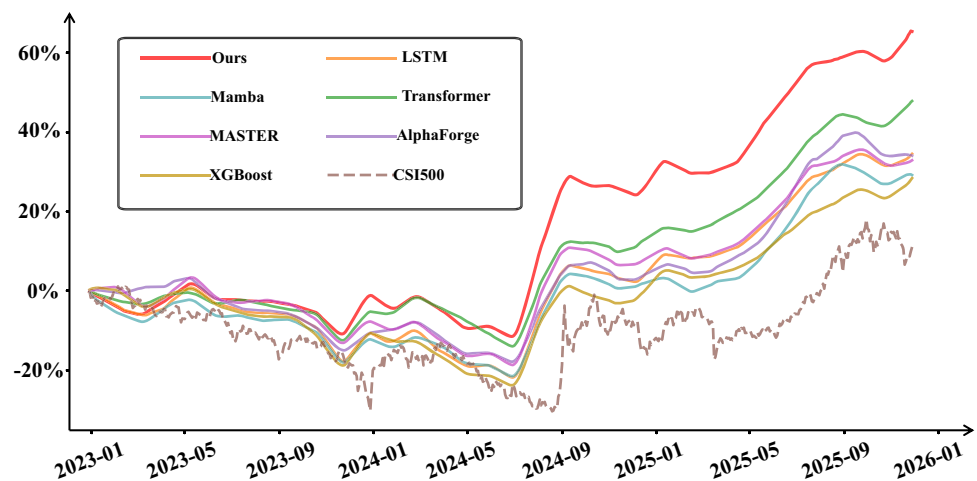


Fig. 3 Visualization of alpha factor scoring and dynamic activation produced by the framework. The y-axis denotes alpha factor indices and the x-axis denotes time steps. **A** Scores assigned to each alpha factor

by MACoE. **B** Scores assigned to each alpha factor by PACoE. **C** Final composite scores determined by the framework. **D** Dynamic activation of alpha factors over time, where “YES” indicates an activated factor

Fig. 4 Simulated trading results: comparison of the cumulative returns of the proposed method with other approaches on the CSI500. The y-axis denotes cumulative return, and the x-axis denotes time in year-month format



tor performance. Figure 3 extends Table 3 by presenting a 50 trading day visualization. Investors can perform a deeper analysis of each alpha factor’s intrinsic meaning and apply that understanding to make informed investment decisions.

Moreover, the model’s interpretability facilitates attribution of return sources and supports the refinement of investment strategies.

Investment simulation

To assess the practical effectiveness of our framework, we conducted simulated trading based on model predictions. The trading rule is as follows: hold at most 50 stocks (selected daily according to model forecasts), limit daily turnover to a maximum of 5 stocks to reduce trading costs, and enforce a minimum holding period of 20 trading days per position to enhance operational feasibility. Additionally, to smooth the return series and simplify calculations, realized returns from a sold position are evenly distributed across its holding period. Figure 4 compares the cumulative returns produced by different models against the CSI500 index. The results indicate that, in the recent simulated period, our framework attains the best overall performance: it substantially reduces downside risk during market drawdowns while delivering appreciable returns during uptrends. This bidirectional risk-return adaptation demonstrates the framework's robust adaptability in volatile markets and provides a practical, reliable basis for real-world investment decisions. This subsection answers Q4.

Related work

Formulaic alpha factors: Formulaic alpha factors are represented as mathematical expressions formed from raw features and operators. In early work, [16] extended the genetic programming algorithm with a rich series of function sets, providing new directions for formulaic alpha factor generation. [16] adopted mutual information as a fitness measure to uncover nonlinear relationships among alpha factors. AlphaEvolve [3] introduced a new family of alphas capable of combining more complex operations. More recently, AlphaGen [30] treats the predictive performance of downstream combination models (e.g., IC) as a reward signal and trains the generator using policy gradient algorithms such as PPO to produce Alpha factor sets that collaboratively enhance overall prediction performance. AlphaForge [24] proposed a two-stage framework for alpha mining and combination that leverages GANs to produce diverse formulaic alpha factors. While AlphaAgent [25] employs LLMs to mine alpha factors and introduces three regularization mechanisms—originality constraints, hypothesis factor alignment, and complexity control—to balance financial interpretability, structural simplicity, and novelty during factor generation, thereby alleviating factor decay. However, existing combination stage methods typically rely on static strategies and therefore fail to jointly account for market volatility and the temporal dynamics of alpha factor performance.

Machine learning-based alpha factors: Rapid advances in deep-learning have spawned new approaches to stock-

return prediction. Early work focused primarily on single stock time series, applying sequence models such as LSTM [10] and Transformer [26], as well as tree-based methods like XGBoost [2]. More recent model designs explicitly account for interactions among stocks: for example, MASTER [15] incorporates a market-guided gating mechanism to capture temporal dependencies intra and inter stock information. Samba [17] employs a bidirectional Mamba [11] block to capture long-term temporal dependencies in stock time series and combines this with graph convolution to uncover cross-stock relationships. Compared with formulaic alpha factors, these machine learning-based approaches generally offer superior predictive flexibility but typically exhibit limited interpretability.

Conclusion

We introduce FactorMoE, a novel framework for combining alpha factors that provides a practical stock-selection tool for investors. FactorMoE introduces CoE, Attention mechanisms, Scoring MoE and gating network. By integrating market regime indicators and alpha factor performance into the factor representations, the framework enables dynamic combination of alpha factors. Through extensive experiments, we show that FactorMoE retains the interpretability of formulaic alpha factors while achieving performance on deep-learning models. In simulated trading, FactorMoE adapts effectively to market volatility and yields superior returns.

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Declarations

Conflict of interest The authors have no conflict of interest to declare that are relevant to the content of this article.

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